

Practical 1: Student Marks Analyzer

Problem Definition

Store student data as a list of tuples. Remove duplicate marks, calculate average marks and store result in a dictionary.

Sample Input

```
students = [  
    (1, "Amit", [78, 85, 90, 85]),  
    (2, "Neha", [88, 92, 79, 88]),  
    (3, "Ravi", [70, 75, 80, 75])  
]
```

Sample Output

```
{1: ('Amit', 84.33), 2: ('Neha', 86.33), 3: ('Ravi', 75.0)}
```

```
In [4]: students = [  
    (1, "Amit", [78, 85, 90, 85]),  
    (2, "Neha", [88, 92, 79, 88]),  
    (3, "Ravi", [70, 75, 80, 75])  
]  
result = {}  
for i,name,marks in students :  
    avg = round(sum(set(marks))/len(set(marks)),2)  
    result[i] = (name,avg)  
print(result)
```

```
{1: ('Amit', 84.33), 2: ('Neha', 86.33), 3: ('Ravi', 75.0)}
```

Practical 2: Inventory Management

Problem Definition

Remove out-of-stock items and calculate total inventory value.

Sample Input

```
inventory = {  
    'pen': (10, 50),  
    'book': (100, 0),  
    'pencil': (5, 100)  
}
```

Sample Output

```
{'pen': (10, 50), 'pencil': (5, 100)}
Total Value: 1000
```

```
In [8]: inventory = {
        'pen': (10, 50),
        'book': (100, 0),
        'pencil': (5, 100)
      }
result = {}
totalValue = 0
for i,(price,qty) in inventory.items():
    if(qty > 0):
        result[i] = (price,qty)
        totalValue+=price*qty
print(result)
print(f"Total Value:{totalValue}")
```

```
{'pen': (10, 50), 'pencil': (5, 100)}
Total Value:1000
```

Practical 3: Email Domain Extractor

Problem Definition

Extract email domains and count occurrences.

Sample Input

```
emails = ['a@gmail.com', 'b@yahoo.com', 'c@gmail.com']
```

Sample Output

```
{'gmail.com': 2, 'yahoo.com': 1}
```

```
In [14]: emails = ['a@gmail.com', 'b@yahoo.com', 'c@gmail.com']
result = {}
for i in emails:
    domain = i.split('@')[1]
    result[domain] = result.get(domain,0) + 1
print(result)
```

```
{'gmail.com': 2, 'yahoo.com': 1}
```

Practical 4: Shopping Cart Billing

Problem Definition

Calculate item-wise cost and total bill.

Sample Input

```
cart = [('apple',2), ('banana',3)]
prices = {'apple':50, 'banana':10}
```

Sample Output

```
{'apple': 100, 'banana': 30}
Total Bill: 130
```

```
In [17]: cart = [('apple',2), ('banana',3)]
prices = {'apple':50, 'banana':10}
result = {}
totalBill = 0
for i,qty in cart:
    result[i] = prices[i]*qty
    totalBill+=prices[i]*qty
print(result)
print(f"Total Bill: {totalBill}")
```

```
{'apple': 100, 'banana': 30}
Total Bill: 130
```

Practical 5: Character Position Index

Problem Definition

Store positions of each character in a string.

Sample Input

```
s = "banana"
```

Sample Output

```
{'b': [0], 'a': [1, 3, 5], 'n': [2, 4]}
```

```
In [22]: s = "banana"
result = {}
for i in range(len(s)):
    if s[i] in result:
        result[s[i]].append(i)
    else :
        result[s[i]] = [i]
print(result)
```

```
{'b': [0], 'a': [1, 3, 5], 'n': [2, 4]}
```

Practical 6: Survey Data Aggregation

Problem Definition

Count responses city-wise and choice-wise.

Sample Input

```
responses = [
    ('Amit', 'Rajkot', 'Yes'),
```

```
( 'Neha', 'Rajkot', 'No' ),  
( 'Ravi', 'Ahmedabad', 'Yes' )  
]
```

Sample Output

```
{ 'Rajkot': { 'Yes': 1, 'No': 1 }, 'Ahmedabad': { 'Yes': 1 } }
```

```
In [1]: responses = [  
    ( 'Amit', 'Rajkot', 'Yes' ),  
    ( 'Neha', 'Rajkot', 'No' ),  
    ( 'Ravi', 'Ahmedabad', 'Yes' )  
]  
result = {}  
for name,city,res in responses :  
    if city not in result :  
        result[city] = {}  
    if res in result[city] :  
        result[city][res] += 1  
    else :  
        result[city][res] = 1  
print(result)
```

```
{ 'Rajkot': { 'Yes': 1, 'No': 1 }, 'Ahmedabad': { 'Yes': 1 } }
```